Practice 1

```
load carsmall;
% cars = importdata('carsmall.mat')
```

What are the mean, standard deviation and median of Acceleration?

```
mean(Acceleration)
```

```
ans = 
15.0280
```

```
std(Acceleration)
```

```
ans = 
3.3485
```

```
median(Acceleration)
```

```
ans = 
15.1500
```

```
%quantile(Acceleration, .5)
```

From MPG (Miles Per Gallon) consumption calculate LPHKM (Liter Per Hundred KiloMeters), if

- 1 gallon = 3.79 liter
- 1 mile = 1.61 kilometers

Hint: element-by-element oprations are

.*	multiplication
./	division
.^	exponentiation

```
LPHKM = 1./(MPG * 1.61 / 3.79)*100;
```

Round it to 1 decimal place:

```
LPKHM = round(LPHKM,1);
```

Which car has the highest consumption? Use the 'Model' character variable to get the name of the car.

```
ind = find(LPKHM == max(LPKHM));
Model(ind,:)
```

```
ans = 
'ih 1200d                         '
```

Count the number of cars with LPKHM > 15 liter. Whats the relative frequency of them?

```
sum(LPKHM > 15)
```

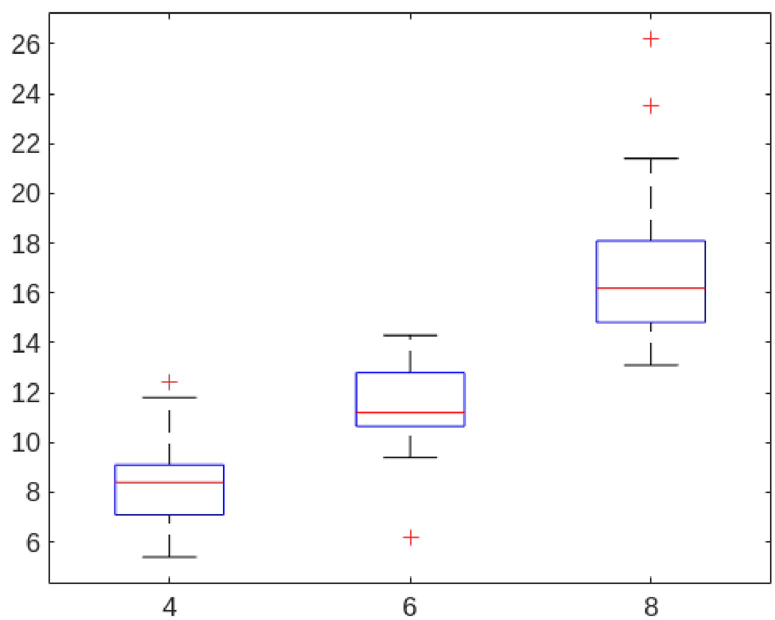
```
ans =
20
```

```
sum(LPKHM > 15) / length(LPKHM)
```

```
ans =
0.2000
```

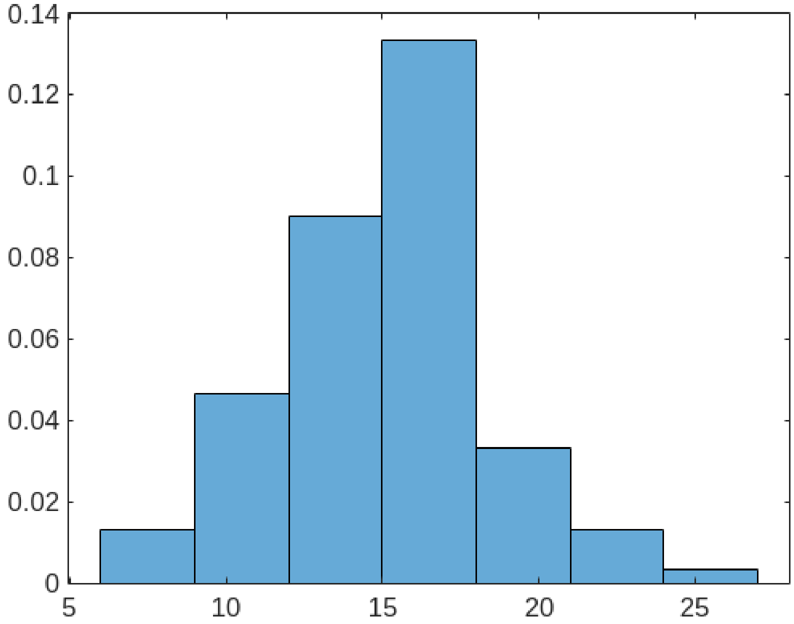
Investigate consumption, conditional to number of cylinders with a boxplot.

```
boxplot(LPKHM, Cylinders)
```

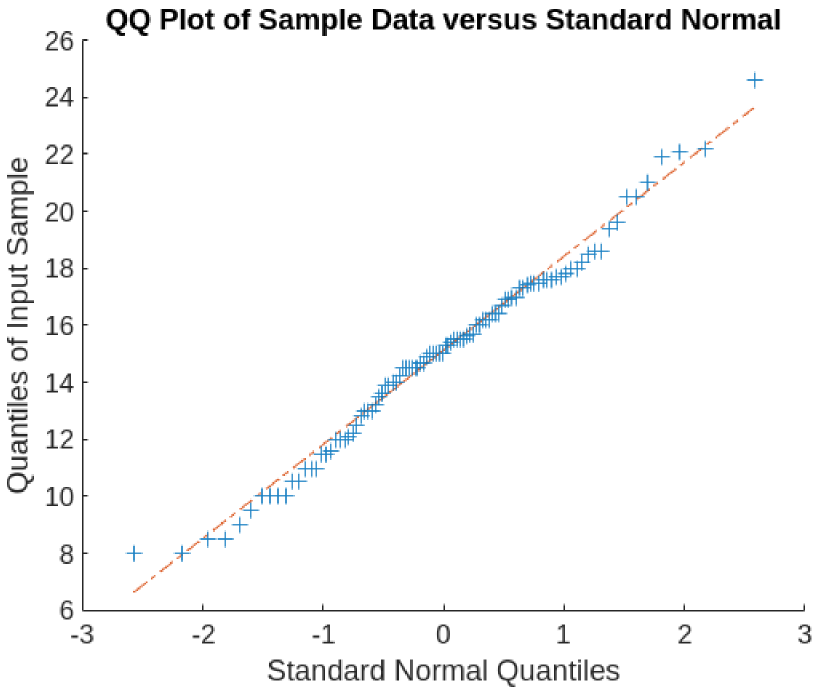


Plot a probability histogram on Acceleration. Does it look normally distributed?

```
histogram(Acceleration,'Normalization','pdf')
```

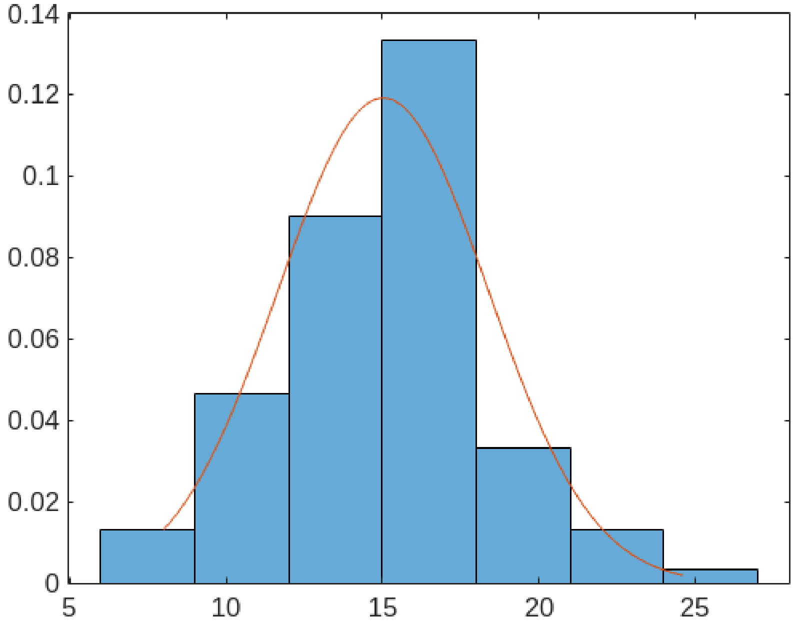



```
qqplot(Acceleration)
```



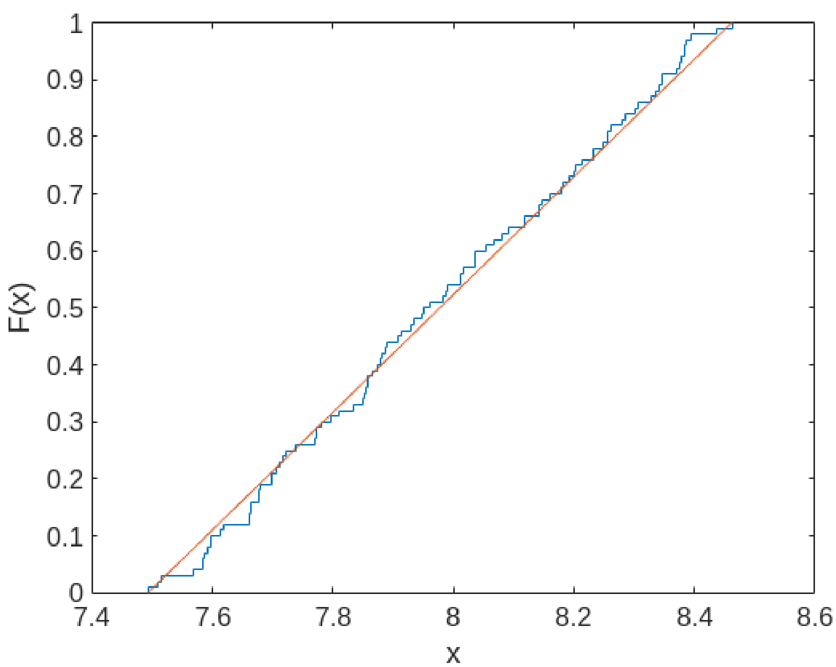
Try to fit a PDF on the histogram. (See also **dfittool**)

```
[pm pd] = normfit(Acceleration); % parameter estimations of the mean and std
xx = linspace(min(Acceleration),max(Acceleration));
histogram(Acceleration,'Normalization','pdf')
hold on;
plot(xx, normpdf(xx,pm,pd))
hold off;
```



Plot the Empirical CDF of log(Weight). Also fit an appropriate theoretical CDF.

```
ecdf(log(Weight))
[a b] = unifit(log(Weight));
xx = linspace(a, b);
hold on;
plot(xx, unifcdf(xx,a,b))
hold off;
```



Make a frequency table from the 'Origin' of cars.

```
tbl = tabulate(Origin);
t = cell2table(tbl,'VariableNames', ...
    {'Value','Count','Percent'});
t.Value = categorical(t.Value)
```

t = 6×3 table

	Value	Count	Percent
1	USA	69	69
2	France	4	4
3	Japan	15	15
4	Germany	9	9
5	Sweden	2	2
6	Italy	1	1

Plot a barplot.

```
bar(t.Value,t.Count)
xlabel('Country of Origin')
ylabel('Number of Cars')
```